

MATTHEW LEE

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EDUCATION

University of California Berkeley

Bachelor of Science in Electrical Engineering and Computer Sciences (EECS)

Berkeley, CA

Expected May 2028

- **GPA:** 4.0/4.0
- **Relevant Coursework:** Linear IC Design; Applied E&M; Computer Architecture; Signals & Systems
- **Accolades:** Dean's List (x3); Member of Eta Kappa Nu (EECS Honor Society)
- **Positions:** Lab TA for EE64 (Intro Circuits); Electronics Admin for Makerspace (Supernode)

EXPERIENCE

Digital Engineer Intern

May 2026 – Aug. 2026

Northrop Grumman Corporation - Space Park

Manhattan Beach, CA

- Conducted design verification and hardware analysis for a **Class A Satellite** program, evaluating board-level power architectures, high-speed **SERDES**, **DDR4** memory, and fiber-optic transceivers.
- Executed **derating analysis** and **SI/PI simulations** to proactively identify 20+ signal and power interface incompatibilities, preventing critical board re-spins downstream.
- **Pioneered** the integration of **Siemens Hyperlynx Schematic Analysis** tools at NGC, streamlining the design-review pipeline and reducing engineering workload.
- Prototyped a custom **Power Evaluation Board** to model and validate **stability** of cascaded **switching regulators**.

Electrical Team Advisor (Former Lead)

Apr. 2025 – Present

UC Berkeley Solar Vehicle Team (CalSol)

Berkeley, CA

- Developed the solar vehicle's **high-voltage (120V Li-ion)** and **low-voltage CAN power system architecture**, integrating a **BMS** and **distribution boards** for safe, efficient, and reliable operation.
- Led hardware and firmware validation for vehicle electronics, implementing multi-layer **BPS fault testing** on **BMS architecture** and **system harnesses** to ensure **zero critical failures** over **1,000+ highway miles**.
- Designed a new **ESP32 firmware** stack supporting **CAN-based communication** between boards, creating a simple and robust architecture for modular development.
- Acted as lead schematic and layout reviewer for **10+ PCBs**, catching **50+ mistakes** before manufacturing

PROJECTS

Two Stage Op-amp LCD Driver

Mar. 2026 — May. 2026

UC Berkeley – EE140 Course Project

Berkeley, CA

- Designed a **two stage** amplifier with a **folded cascode** first stage, **class AB Monticelli** biased second stage, and an **Ahuja Compensation** feedback to meet design specifications and create a **PVT insensitive** conscience design.
- Optimized power and settling time through **transistor sizing**, using an iterative workflow between **gm/Id Python** LUT scripts and custom **Cadence Maestro** workbenches.
- Performed **return ratio analysis** and **two-port modeling** to inform topology choices and **pole placement**.

Audio PCB for Car Horn

Oct. 2024 – May. 2025

UC Berkeley Solar Vehicle Team (CalSol)

Berkeley, CA

- Implemented an **I2C-based DAC**, **bandpass filtering**, and **high-gain amplification** to optimize **horn signal integrity** and minimize **power loss**.
- Engineered a **dual-voltage power system** featuring a **FET priority ORing circuit** (18V primary / 12V CAN backup) and a **3.3V buck regulator** to reliably power an ESP32 host and **high-current power amplifier stage**.
- Routed USB/UART **differential pairs** and **high-current power polygons**, successfully resolving thermal trace necking and UART pinout faults.

TECHNICAL SKILLS

Tools & Platforms: KiCAD, Siemens Xpedition, Siemens Hyperlynx, SPICE, Cadence Virtuoso, Cadence Maestro, Platform.io, VS Code, Git, Confluence, SolidWorks, Microsoft Suite, Jira

Hardware: PCB Design & Layout, Mixed-signal Circuit Design, Power Electronics, Filters, Small Signal

Embedded Systems & Languages: Embedded C/C++, RISCv Assembly, ESP32, CAN, SERDES, SPI, I2C, UART, Python

Validation & Testing: PCB bring-up, System-level verification and integration, Documentation, BUS Monitoring, Harnessing, Soldering, Oscilloscope Testing, Logic Analyzer